

Daily CCS Exec Briefing — 18 August 2026

For Ayu · Geologist, PETRONAS CCS Track

Compiled 13:28 MYT · Source: memory intel + confirmed signal patterns

Live news tickers not reachable (SearXNG down, DuckDuckGo timeout, search empty)

⚠ **Source caveat:** Search backends all returned empty today (SearXNG empty, DuckDuckGo timeout, Google blocked). This brief combines locked memory intel with known-pattern signals — not fresh tickers. Items flagged "verify" need Ayu's own cross-check via Reuters Energy / Upstream / Energy Voice Asia-Pacific desk.

1. PETRONAS Kasawari CCS — Status Track

- **FID announced** (2023), target storage ~3.3 Mtpa CO₂ in depleted **M4 carbonate (offshore Sarawak)**
- **Injection start window:** public expectation 2025–2026. Exact start date not locked in memory — Ayu, who is in the project, knows the operational reality better than this brief does
- **Geological read:** depleted gas reservoir = proven trap integrity, well-characterized pressure-depletion history = baseline for re-pressurization modeling. CO₂ injection will induce pressure buildup → **caprock fault reactivation & seal breach** is the primary risk. Dynamic geomechanics + 4D seismic monitoring baseline are mandatory
- **Signals to watch:** well integrity test reports, microseismic monitoring data, plume tracking via time-lapse seismic, halokinesis effect on M4 caprock

2. Malaysia CCS Regulatory — CCUS Bill

- Range of motion: Malaysia moving toward dedicated CCUS regulatory framework. Specific Act for CCS storage liability & permitting still in drafting phase (per memory intel as of 2024–2025)
- **Ayu-relevant:** if handling subsurface characterization for storage permit applications, regulatory uncertainty around long-term liability (30+ years post-closure) and transfer of liability from operator to state is the critical area. **PSL (post-storage liability) framework — who holds it?**

- **Cross-check needed:** actual status of CCUS Bill in Parliament — confirmation tickers not reachable today

3. Southeast Asia CCS Geopolitics — Pattern

- **Indonesia CCS hub push** — Indonesia aggressively offering CCS-as-service, attracting Western majors. Reuters/Bloomberg reports not reachable live for verification today
- **Singapore–CCS chain** — cross-border CO₂ transport from Singapore to Malaysia/Indonesia storage sites still in study phase
- **Implication for PETRONAS:** Sarawak storage advantage (depleted fields, deepwater proximity to high-emission East Asia) = strategic asset. But competitor entry early = pricing pressure downstream

4. Geoscience Technical Pulse — What CCS Geologists Must Watch

1. **Injectivity vs capacity** — pressure buildup modeling & fracture gradient.
Reservoir quality (porosity/permeability) of M4 carbonate is baseline-critical data
2. **Caprock integrity** — seal capacity under re-pressurization. M4 caprock shale — what shale gouge ratio, what capillary entry pressure?
3. **Plume migration prediction** — reactive transport simulation (CO₂–brine–rock).
Mineral trapping potential in limestone vs sandstone?
4. **Well integrity legacy** — old wells through caprock become leakage pathways.
P&A status, cement bond logs, sustained casing pressure tests
5. **Monitoring protocol** — InSAR, 4D seismic, microseismic, fluid sampling.
Frequency dictated by regulator — and regulator is still figuring it out
6. **MRV (Monitoring, Reporting, Verification)** — IPCC-tier accounting. Malaysia needs to develop MOC (margin of conservatism) framework

5. Energy Transition Signal — Macro

- **Carbon pricing** — Bursa Carbon Exchange active since Dec 2022. Prices volatile.
If Ayu handles CCS for voluntary carbon market — verify credit methodology (Verra / Plan Vivo / PETRONAS internal)
- **Hydrogen–CCS nexus** — blue hydrogen requires CCS as critical dependency.
Japan/Korea off-takers drive this chain. Sarawak H₂ hub depends on Kasawari-class storage

- **Methane abatement vs CCS** — short-term cheaper to plug methane leaks than deploy CCS. But structural decarbonization needs CCS

⚠ **What I Cannot Verify Live Today**

- Actual Kasawari injection start date (if started — breakthrough vs behind schedule?)
- CCUS Bill Parliament status (2nd reading? passed?)
- Competitor deals (Indonesia–Pertamina CCS hub progress, Singapore 1.5°C-aligned strategies)
- Bursa Carbon Exchange spot carbon price today
- Any major incident at global CCS sites (leak incidents, well blow-outs, public opposition)

Saran: cross-check with Reuters Energy, Upstream, Energy Voice — their Asia-Pacific CCS coverage is freshest. If handling public communication on Kasawari, any PETRONAS press release this morning may have slipped past me.

Bottom Line For Ayu

1. **If handling subsurface workstream:** your focus right now is caprock integrity modeling + legacy well audit. These two are the global industry bottleneck
2. **If handling stakeholder communication:** be ready — Malaysia CCS regulatory uncertainty is still real. Every project is bespoke negotiation
3. **If new to CCS:** monitor global incidents (Quest CCS Canada, Gorgon Australia, Northern Lights Norway) — their post-mortem published reports are all available. That's your core curriculum

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